

CO223_Lab02

E/21/019

1. Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?

```
▼ Hypertext Transfer Protocol
  ▶ GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
```

2. What languages (if any) does your browser indicate that it can accept to the server?

```
Accept-Language: en-US,en;q=0.9\r\n
```

3. What is the IP address of your computer? Of the gaia.cs.umass.edu server?

```
Source Address: 10.40.18.129
Destination Address: 128.119.245.12
```

4. What is the status code returned from the server to your browser?

```
Response Version: HTTP/1.1
Status Code: 200
```

5. When was the HTML file that you are retrieving last modified at the server?

```
Last-Modified: Fri, 13 Dec 2024 06:59:01 GMT\r\n
```

6. How many bytes of content are being returned to your browser?

```
▼ Content-Length: 128\r\n
  [Content length: 128]
```

7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.

- No

8. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

- No

9. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

847	16:00:28.581379	10.40.18.129	128.119.245.12	HTTP	526 GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
876	16:00:28.903762	128.119.245.12	10.40.18.129	HTTP	784 HTTP/1.1 200 OK (text/html)

- The file content was returned. (status code : 200 OK)

10. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

```
If-Modified-Since: Fri, 13 Dec 2024 06:59:01 GMT\r\n\r\n
```

1457	16:00:47.531796	10.40.18.129	128.119.245.12	HTTP	638 GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
1470	16:00:47.848875	128.119.245.12	10.40.18.129	HTTP	294 HTTP/1.1 304 Not Modified

11. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

```
Status Code: 304  
[Status Code Description: Not Modified]  
Response Phrase: Not Modified
```

12. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill of Rights?

684	16:17:53.647960	10.40.18.129	128.119.245.12	HTTP	526 GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
709	16:17:53.935464	128.119.245.12	10.40.18.129	HTTP	535 HTTP/1.1 200 OK (text/html)

684

13. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

709

14. What is the status code and phrase in the response?

```

HTTP/1.1 200 OK\r\n
  Response Version: HTTP/1.1
  Status Code: 200
  [Status Code Description: OK]
  Response Phrase: OK

```

15. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?

```

TCP segment data (481 bytes)
[2 Reassembled TCP Segments (4861 bytes): #708(4380), #709(481)]

```

16. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

- Three messages are present. The web addresses are displayed below.

673	16:31:19.534537	10.40.18.129	128.119.245.12	HTTP	526 GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
709	16:31:19.856703	128.119.245.12	10.40.18.129	HTTP	1355 HTTP/1.1 200 OK (text/html)
717	16:31:19.878578	10.40.18.129	128.119.245.12	HTTP	472 GET /pearson.png HTTP/1.1
734	16:31:20.088491	10.40.18.129	178.79.137.164	HTTP	439 GET /8E_cover_small.jpg HTTP/1.1
737	16:31:20.175138	128.119.245.12	10.40.18.129	HTTP	745 HTTP/1.1 200 OK (PNG)
743	16:31:20.284808	178.79.137.164	10.40.18.129	HTTP	225 HTTP/1.1 301 Moved Permanently

17. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain

- It's obviously a parallel download because the time difference is so tiny milliseconds or even fractions of a second.

717	16:31:19.878578	10.40.18.129	128.119.245.12	HTTP	472 GET /pearson.png HTTP/1.1
734	16:31:20.088491	10.40.18.129	178.79.137.164	HTTP	439 GET /8E_cover_small.jpg HTTP/1.1

18. What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?

```

Status Code: 401
[Status Code Description: Unauthorized]
Response Phrase: Authorization Required

```

19. When your browser's sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?

```
▼ Authorization: Basic d2lyZXNoYXJrLXN0dWR1bnRzOm5ldHdvcm5=\r\n  
  Credentials: wireshark-students:network
```